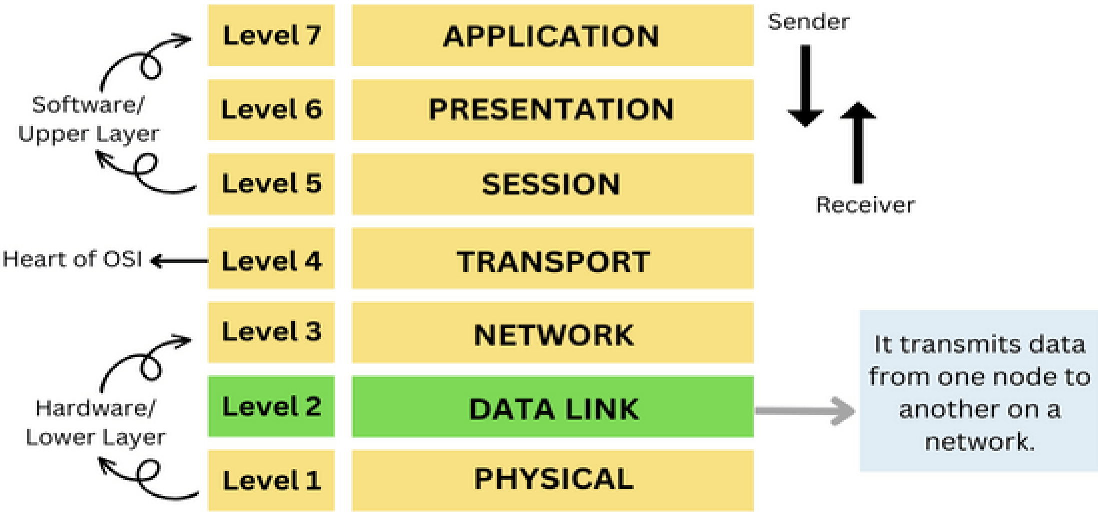
UNIT :02

Data Link Layer and Medium Access Sub Layer: Error Detection and Error Correction - Fundamentals, Block coding, Hamming Distance, CRC; Flow Control and Error control protocols - Stop and Wait, Go back – N ARQ, Selective Repeat ARQ, Sliding Window, Piggybacking, Random Access, Multiple access protocols -Pure ALOHA, Slotted ALOHA, CSMA/CD,CDMA/CA.

Data Link Layer :

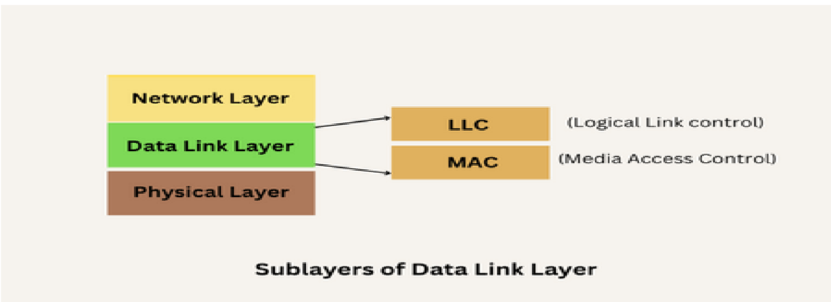
- The data link layer is the second layer from the bottom in the OSI (Open System Interconnection) network architecture model.
- It is responsible for the **node-to-node delivery of data**.
- Its major role is to ensure **error-free transmission** of information.
- DLL is also responsible for **encoding, decoding, and organizing the outgoing and incoming data.**
- In this article, we will discuss Data Link Layer in Detail along with its functions, and sub-layers.



The OSI Model: Data Link Layer

❖ **Sub-Layers of The Data Link Layer :**

The data link layer is further divided into two sub-layers, which are as follows:

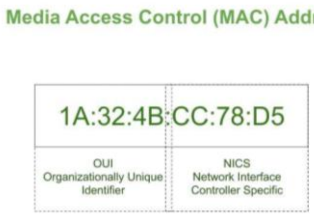


1. **Logical Link Control (LLC) or Data Link Control (DLC) Sublayer:**

- LLC or DLC is the topmost layer of the data link layer.

- It deals with the communication between the lower layers and upper layers.
- It Handles error detection, flow control, and framing.
- Provides an interface to the **Network Layer.**
- Independent of the network hardware.
- It is responsible for assigning the frame sequence number.
- This sublayer of the data link layer deals with multiplexing, the flow of data among applications and other services .

2. Media Access Control (MAC) :



- The bottom sublayer of the Data Link Layer is the Media Access Control. **(Lower part of the DLL.)**
- It is also known as Medium Access Control.
- Manages how devices access the physical medium (e.g., Ethernet, Wi-Fi).
- Deals with **physical addressing** and controls how the medium is accessed.
- Ensures efficient use of shared communication media.
- It provides multiplexing and flow control for the transmission media.
- It determines who is permitted to access the media at any given time.

❖ Framing

- Converts a stream of bits from the Physical Layer into frames with headers and trailers.
- **Common methods:**

1. **Character Count:**
   - Includes a count of characters in the header.
   - Issue: If the count field is corrupted, the entire frame becomes unreadable.
2. **Flag Bytes with Byte Stuffing:**
   - Special **flag bytes** (e.g., `01111110`) are used to mark the beginning and end of the frame.
   - Extra bytes (stuffed bytes) are added if flags appear in the data.
3. **Bit Stuffing:**
   - Avoids ambiguity in the data by inserting **extra bits** when flag patterns are detected.

❖ Error Handling in Data Communication

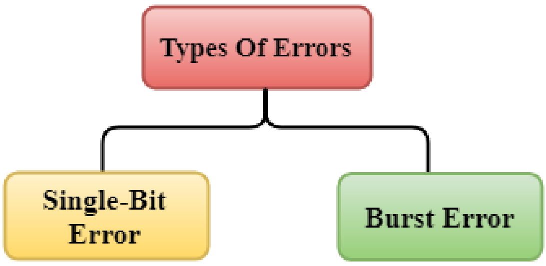
What is an Error?

- An **error** occurs when the received message differs from the transmitted message.
- **Error detection** techniques identify whether errors have occurred during transmission, without correcting them.

Types Of Errors :

Errors can be classified into two categories:

1. Single-Bit Error
2. Burst Error

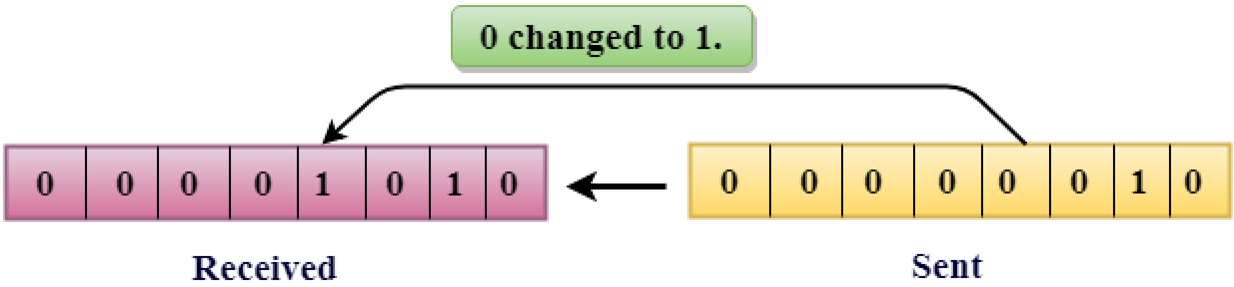


1. Single-Bit Error:

The only one bit of a given data unit is changed from 1 to 0 or from 0 to 1.

For example:

- Sent: 00000010
- Received: 00001010



In the above figure, the message which is sent is corrupted as single-bit, i.e., 0 bit is changed to 1.

- **Single-Bit Error** does not appear more likely in Serial Data Transmission.
- Single-Bit Error mainly occurs in Parallel Data Transmission.

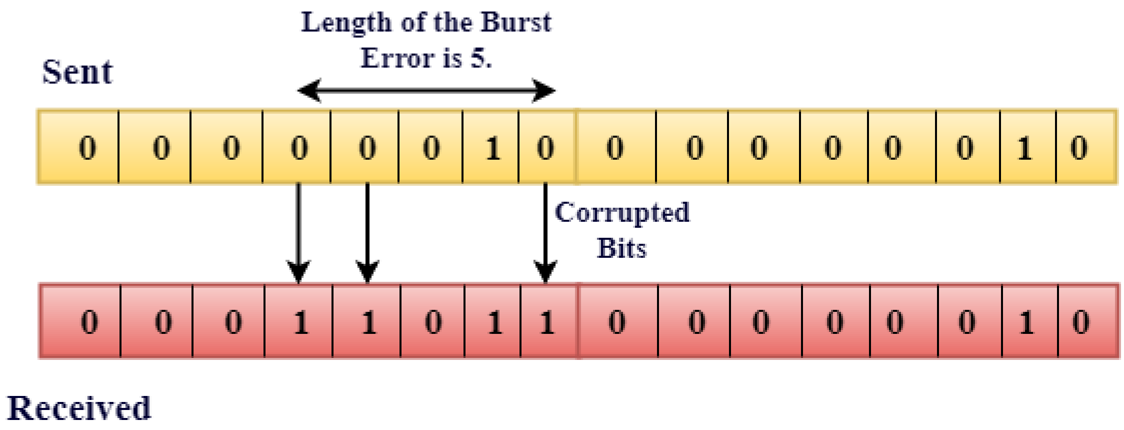
2. **Burst Error :**

- The two or more bits are changed from 0 to 1 or from 1 to 0 is known as Burst Error.
- The Burst Error is determined from the first corrupted bit to the last corrupted bit
- Occurs when two or more bits in the data unit are altered during transmission.

- **Two or more bits are altered.**
- **Length is measured from the first corrupted bit to the last.**

For example:

- Sent: 0000001000000010
- Received: 0001101100000010



- **Error Detecting Techniques:**

Error-detecting techniques are used in data communication to identify errors that may occur during data transmission.

Here's a brief overview of the most popular Error Detecting Techniques are:

1. Single parity check
2. Two-dimensional parity check
3. Checksum
4. Cyclic redundancy check

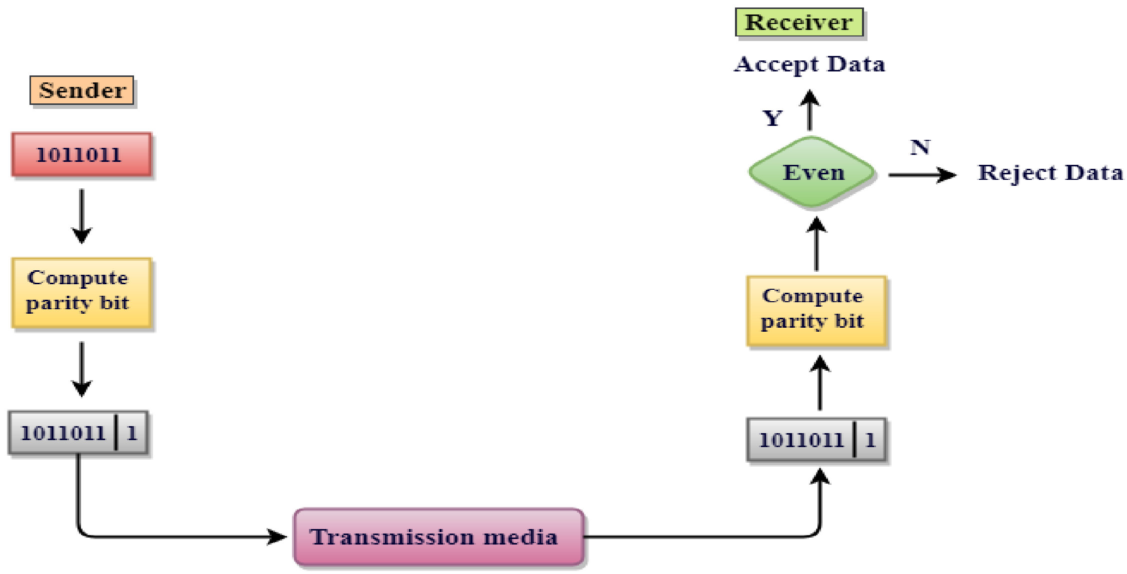
1. Single Parity Check :

- **Single Parity Checking** is a simple and cost-effective error detection mechanism.
- It is widely used in systems where minimal error detection capability is acceptable.

Working:

- For **even parity**, the parity bit is set to 1 if the total number of 1's in the data is odd (to make it even)
- For **odd parity**, the parity bit is set to 1 if the total number of 1's in the data is even (to make it odd).

➢ This technique generates the total number of 1's even, so it is known as **even-parity checking.**

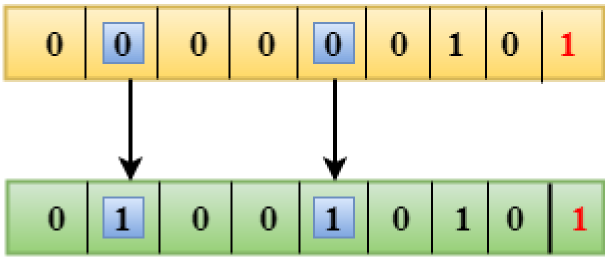


Advantage:

- Simple and easy to implement.
- Simple parity check can detect all single bit error.
- Simple parity check can detect an odd number of errors.

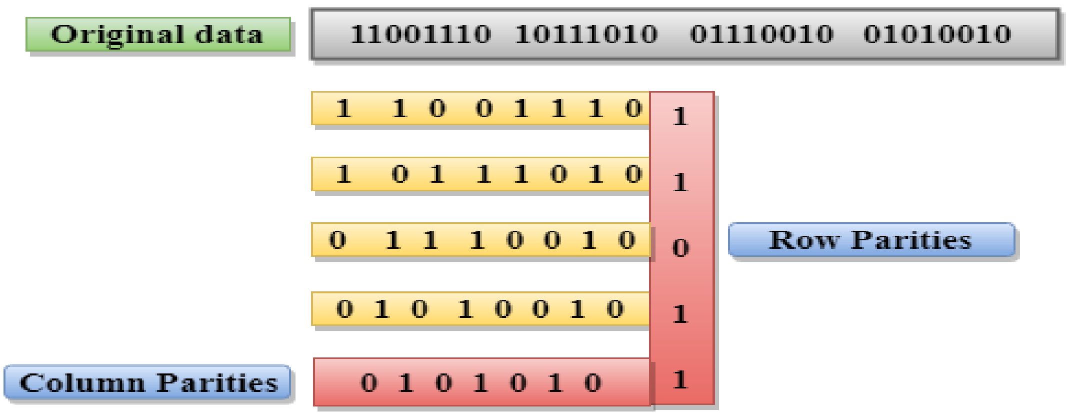
Drawbacks Of Single Parity Checking

- It can only detect single-bit errors which are very rare.
- If two bits are interchanged, then it cannot detect the errors.



2. Two-Dimensional Parity Check

- Performance can be improved by using **Two-Dimensional Parity Check** which organizes the data in the form of a table.
- Data is arranged in a table.
- In Two-Dimensional Parity check, a block of bits is divided into rows, and the redundant row of bits is added to the whole block.
- At the receiving end, the parity bits are compared with the parity bits computed from the received data.
- Row and column parity bits are calculated for error detection.



3. Checksum :

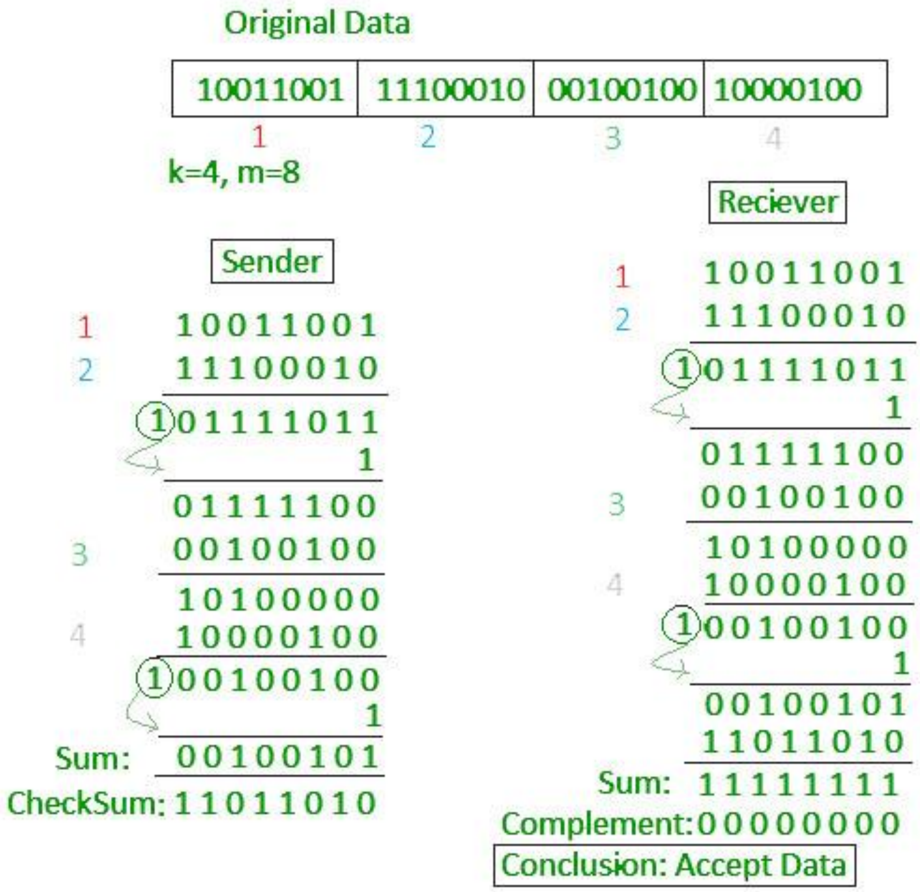
- A Checksum is an error detection technique based on the concept of redundancy.
- Data is divided into equal-sized segments, and their binary sum is calculated.
- The calculated sum is then sent along with the data to the receiver.
- The receiver recalculates the checksum to verify integrity.

Checksum – Operation at Sender's Side

- Firstly, the data is divided into k segments each of m bits.
- On the sender's end, the segments are added using 1's complement arithmetic to get the sum. The sum is complemented to get the checksum.
- The checksum segment is sent along with the data segments.

Checksum – Operation at Receiver's Side

- At the receiver's end, all received segments are added using 1's complement arithmetic to get the sum. The sum is complemented.
- If the result is zero, the received data is accepted; otherwise discarded.

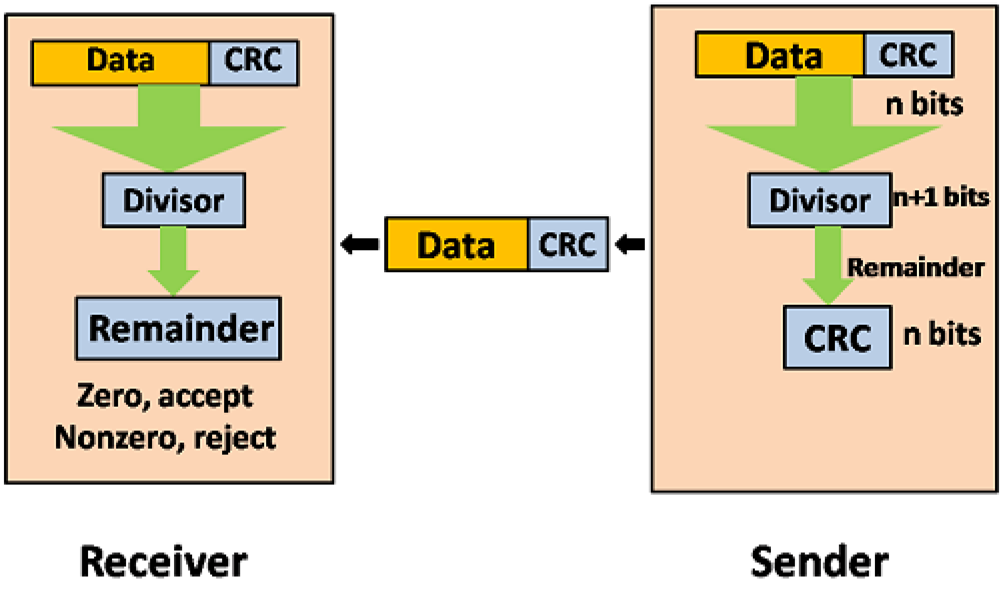


- **Advantage**: Can detect many types of errors.
- **Limitation**: Less effective for detecting burst errors.

4. Cyclic Redundancy Check (CRC):

- A powerful error-detecting technique using polynomial division.
- Data is treated as a binary number or polynomial.
- A polynomial division is performed on the data, and the remainder is sent as the CRC code.
- If the resultant of this division is **zero** which means that it has **no error**, and the data is accepted.

- If the resultant of this division **is not zero** which means that the data consists of **an error**. Therefore, the data is discarded.
- Receiver performs the same division and checks the remainder.
- **Advantage:** Very reliable for detecting burst errors.
- Widely used in networks and storage devices.
- **Limitation:** Slightly more complex to implement than parity checks.



❖ CRC Working

We have given data word of length n and divisor of length k.

Step 1: Append (k-1) zero's to the original message

Step 2: Perform modulo 2 division

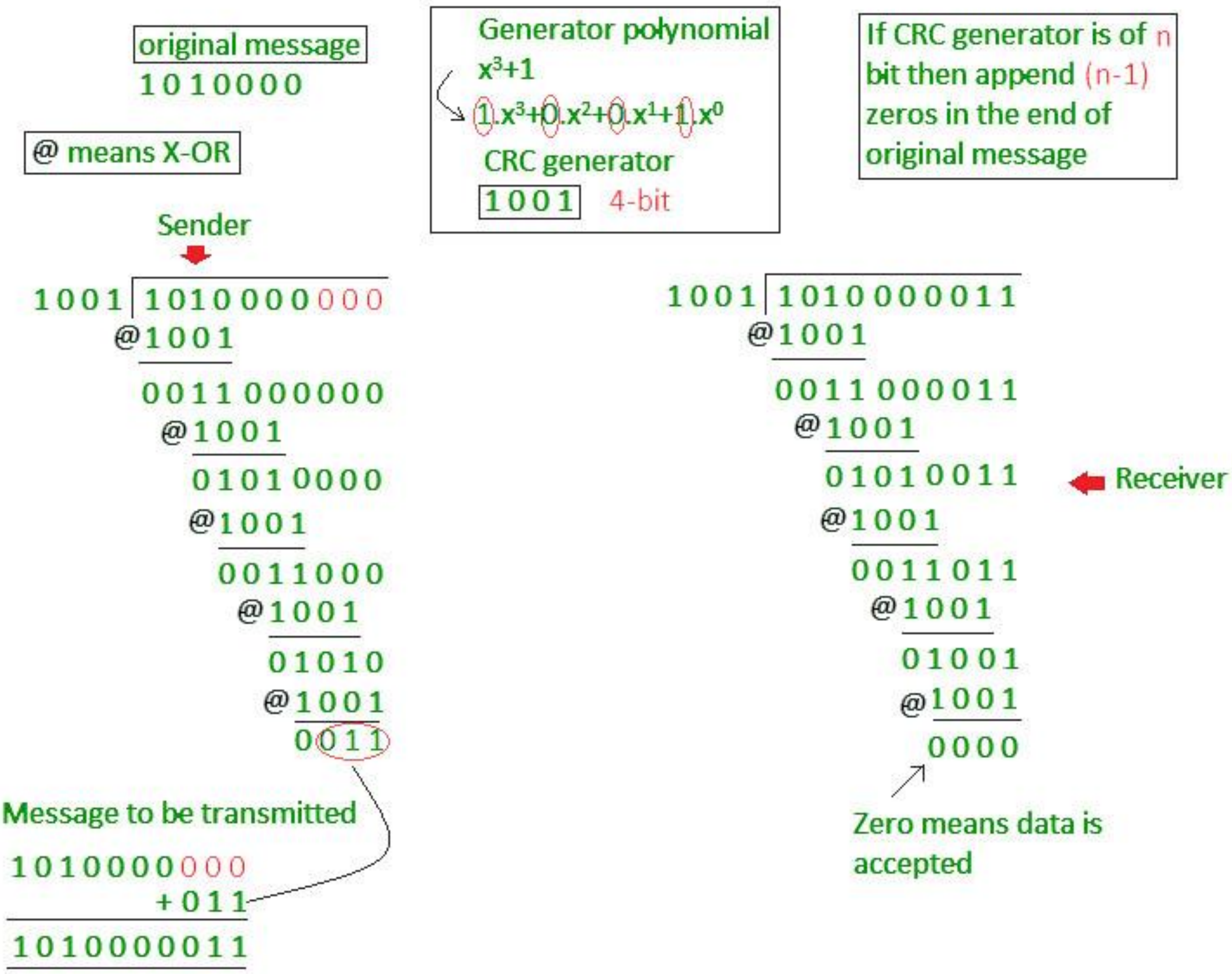
Step 3: Remainder of division = CRC

Step 4: Code word = Data with append k-1 zero's + CRC

Note:

- CRC must be k-1 bits
- Length of Code word = n+k-1 bits

Example: Let's data to be send is 1010000 and divisor in the form of polynomial is x3+1. CRC method discussed below.



- **Error Correction**
  - Error Correction codes are used to detect and correct the errors when data is transmitted from the sender to the receiver.
  - Error Correction can be handled in two ways:

I. **Forward Error Correction:** In this Error Correction Scenario, the receiving end is responsible for correcting the network error. There is no need for retransmission of the data from the sender's side.

II. **Backward Error Correction:** the sender is responsible for retransmitting the data if errors are detected by the receiver. The receiver signals the sender to resend the corrupted data or the entire message to ensure accurate delivery.

❖ Block Coding :

- Block coding is a data encoding method where data is divided into fixed-size blocks, and redundant bits are added to each block.
- These redundant bits help in detecting or correcting errors during data transmission.

❖ Process:

1. **Divide Data**: Split data into k-bit blocks.
2. **Add Redundancy**: Append n −k redundant bits to create n-bit codewords.
3. **Transmit Codeword**: Send the n-bit codeword.
4. **Error Handling**: Use redundant bits to detect or correct errors.

❖ Applications:

- **Error Detection:** Parity bits, checksums.
- **Error Correction:** Hamming codes, cyclic codes.

Example of Block Coding:

Hamming (7, 4) Code:

- **Data block (k = 4):** 1011
- **Redundant bits (n−k=3):** Added based on parity positions (powers of 2).
- **Codeword (n=7):** 1011010
  - 7-bit codeword contains 4 data bits and 3 redundant bits.

Advantages of Block Coding:

- Enables error detection and correction.
- Increases the reliability of data communication.
- Supports various error-detecting and correcting algorithms.

❖ Data Link Controls

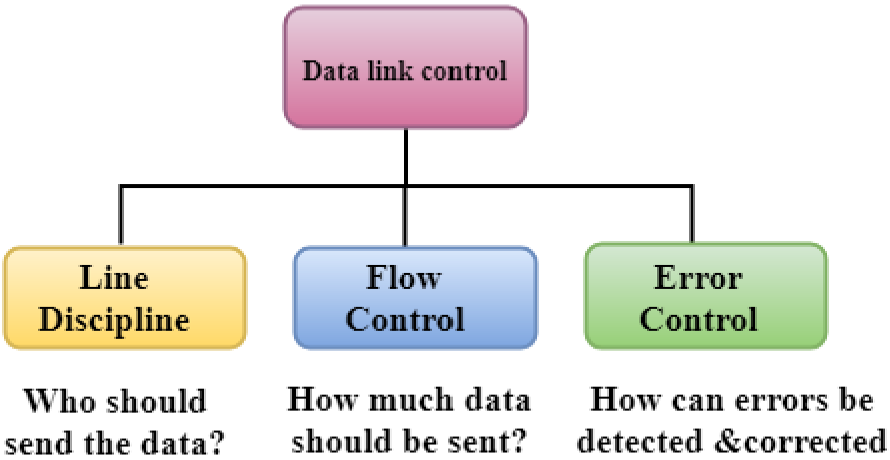
- Data Link Control refers to the techniques and protocols provided by the **Data Link Layer** of the OSI model to ensure **reliable data transfer** over the physical medium.
- This layer manages data framing, error detection, error correction, flow control, and coordination between devices to avoid collisions.

Key Role:
The data link layer ensures that frames are delivered error-free and in the correct sequence.

Example Use Case:
In **half-duplex mode**, where only one device can transmit at a time, the Data Link Layer ensures that both devices coordinate to prevent collisions when they try to send data simultaneously.

The Data link layer provides three functions:

- Line discipline
- Flow Control
- Error Control



1. Line Discipline :

- Line discipline is responsible for **managing the link** and determining **which device sends data** and **when** to send it.
- This ensures that devices communicate in an organized manner.

2. Flow Control :

- It is a technique that generally observes the proper flow of data from sender to receiver.
- It is very essential because it is possible for sender to transmit data or information at very fast rate and hence receiver can receive this information and process it.
- This can happen only if receiver has very high load of traffic as compared to sender, or if receiver has power of processing less as compared to sender.

Two methods have been developed to control the flow of data:

I. **Stop-and-Wait Protocol:**
   - The sender transmits **one frame at a time** and waits for an acknowledgment (ACK) before sending the next frame.
   - In the Stop-and-wait method, the sender waits for an acknowledgement after every frame it sends.

How It Works:

- The sender sends a data frame.
- The sender stops and waits for the receiver to acknowledge receipt.
- Upon receiving the acknowledgment, the sender sends the next frame.
- If no acknowledgment is received (due to an error or loss), the sender retransmits the frame after a timeout.
- **Advantages:**
    - Simple to implement.
    - Reliable for error detection and flow control.
- **Disadvantages:**
    1. Inefficient for high-latency networks
    2. The transmission medium remains idle while waiting for acknowledgment.

II. **Sliding Window Protocol**:

- The sender is allowed to transmit **multiple frames** (up to a window size) before needing acknowledgment.
- The receiver sends ACKs as it processes frames, allowing the sender to transmit more data.

How It Works:

1. The sender maintains a "sliding window" of sequence numbers for frames that can be sent.
2. After sending a frame, the sender doesn't wait for an acknowledgment but continues sending up to the window size.
3. The receiver acknowledges the received frames.
4. Once acknowledgment for a frame is received, the sender "slides" the window forward, allowing new frames to be sent.

Advantages:

- Efficient for high-speed networks.
- Reduces idle time.

Disadvantages:

- Complex to implement compared to Stop-and-Wait.

3. Error Control :

- **Error Control** refers to the mechanisms implemented by the **Data Link Layer** to ensure that data is delivered **error-free**, in the correct order, and without loss or duplication.
- The primary goal is to detect and correct errors that occur during the transmission of data over the communication channel.

❖ Error Control Mechanisms

1. **Error Detection**: The Data Link Layer uses various techniques to detect errors in transmitted data:

- Detects errors in frames using methods like:

I. **Parity Check:**

- A parity bit is added to the data to make the total number of 1s either **even** (even parity) or **odd** (odd parity).
- If the parity is incorrect at the receiver, an error is detected.
- **Limitation:** Cannot detect burst errors.

II. **Checksum:**

- The sender calculates a checksum value (using summation or a similar method) for the data and appends it to the frame.
- The receiver recalculates the checksum for the received data.
- If the values don't match, an error is detected.

III. **Cyclic Redundancy Check (CRC):**

- A **polynomial division-based method** where the sender appends a CRC code to the data.
- At the receiver, the frame is divided by the same generator polynomial.
- If the remainder is non-zero, it indicates an error.

2. **Error Correction**: Once an error is detected, the Data Link Layer employs methods to correct the error.

- Corrects detected errors using methods like:

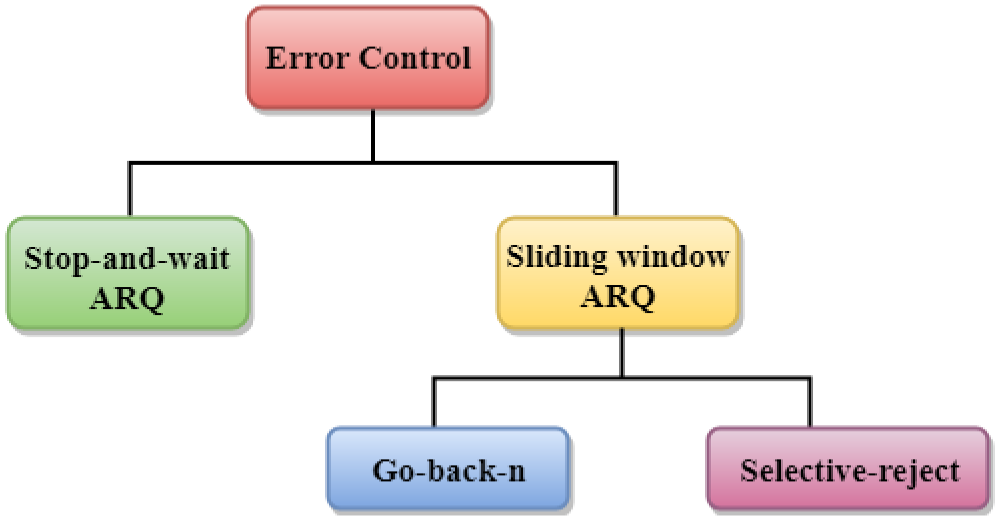
I. **Forward Error Correction (FEC):**

- Redundant bits are added to the frame so that the receiver can identify and correct errors without retransmission.
- Example: **Hamming Code**.

II. **Automatic Repeat Request (ARQ):**

- Relies on retransmissions for error correction.
- If an error is detected, the receiver requests the sender to retransmit the affected frame.

Types of Error Control :



1. **Stop-and-Wait ARQ:**

   - Stop-and-Wait ARQ, also called the **alternating bit protocol**,
   - It is a simple and widely used error control mechanism.
   - It ensures that data is transmitted reliably between two connected devices by alternating between sending data and waiting for acknowledgment.
   - Sender waits for ACK after each frame.
   - Simple but inefficient for long-distance or high-speed link

   Advantages:

   - Simple to implement
   - Ensures reliable delivery of each frame

   Disadvantages:

   - Low efficiency, as only one frame can be sent at a time.
   - Poor bandwidth utilization in high-latency networks.

2. **Sliding Window ARQ** :

   - The **Sliding Window ARQ (Automatic Repeat Request)** is an efficient error and flow control protocol used in data communication to manage the transmission of multiple frames simultaneously.

❖ **How Sliding Window ARQ Works:**

i. **Window Size:**

   - A "window" is defined as the number of frames that the sender is allowed to send without waiting for an acknowledgment.
   - The sender and receiver both maintain a window of a fixed size.

- For a window size of N, the sender can send up to N frames before needing an acknowledgment.

ii. **Sliding Mechanism:**

- After the sender transmits frames within the window size, it waits for an acknowledgment (ACK).
- As the receiver sends ACKs, the sender's window slides forward, allowing it to send more frames.

iii. **Receiver Window:**

- The receiver also maintains a window to track the frames it expects.
- If a frame is received out of order, it may store it temporarily or discard it, depending on the protocol variation (e.g., Go-Back-N or Selective Repeat).

Two Main Sliding Window ARQ:

I. **Go-Back-N ARQ:**

Go-Back-N ARQ (Automatic Repeat Request) is a sliding window protocol used for error control in data communication. It retransmits all frames from the point of error detection or missing acknowledgment, ensuring reliable delivery of data frames.

❖ **Working Principle of Go-Back-N ARQ:**

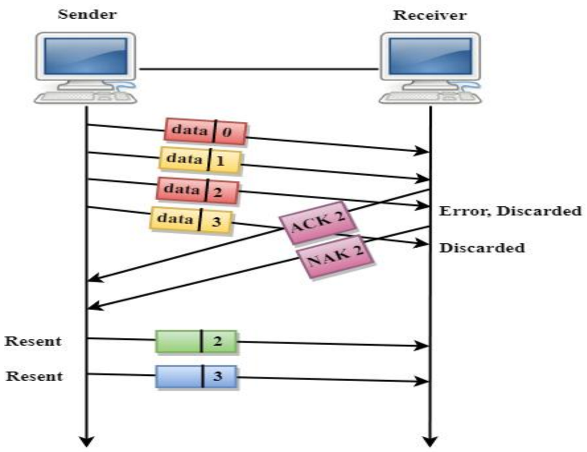
a. The **sender** can send multiple frames without waiting for an acknowledgment but only up to a certain number defined by the window size (N).

b. Each frame is assigned a **sequence number**.

c. The **receiver** sends an acknowledgment (ACK) for every frame it successfully receives.

d. If an error occurs or a frame/ACK is lost:

- The receiver informs the sender by sending a **negative acknowledgment (NAK)** or nothing at all (detected through timeouts).
- The sender then retransmits the erroneous frame **and all subsequent frames**, regardless of whether those subsequent frames were received correctly.



II. **Selective Repeat ARQ:**

Selective-Reject ARQ is an advanced error-control technique that retransmits only the specific frames that were received in error, making it more efficient than Go-Back-N ARQ.

How It Works:

a. **Transmission:** Sender transmits frames within the allowed window size.

b. **Error Detection:** Receiver sends:
   - ACK for correct frames.
   - NAK for erroneous frames.

c. **Retransmission:** Sender retransmits only the frames with errors.

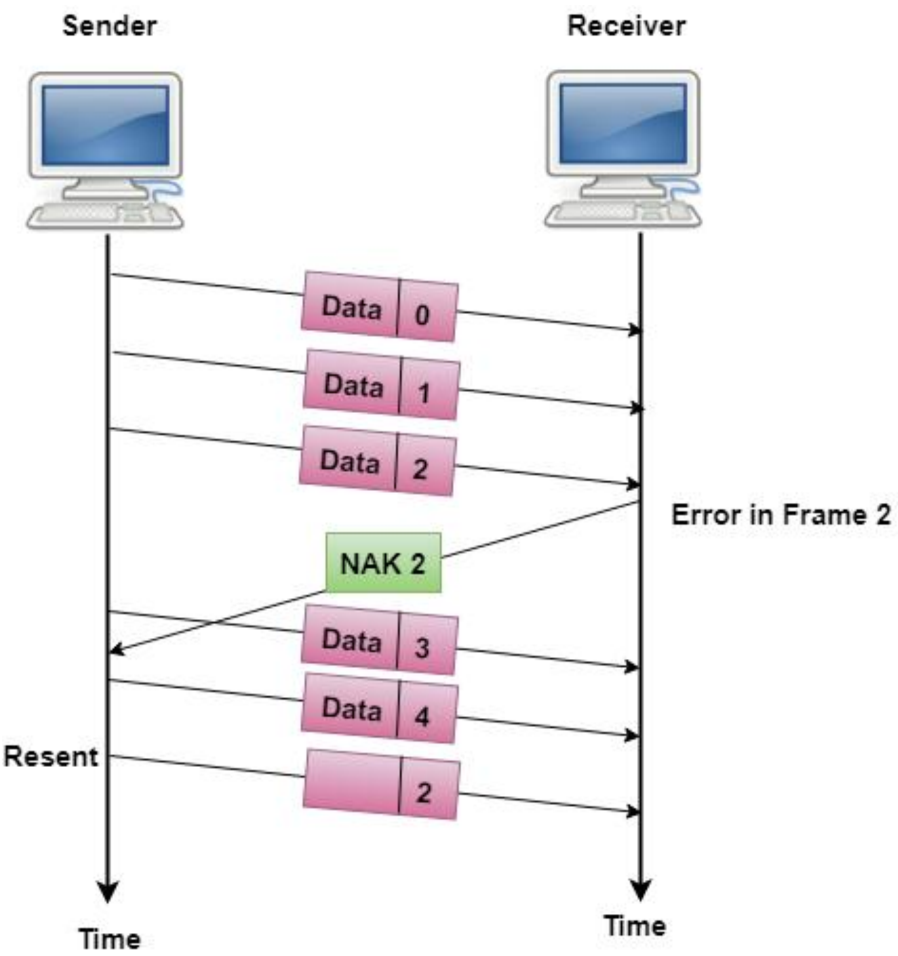
d. **Reordering:** Receiver stores out-of-order frames in a buffer and reorders them after receiving the missing frames.

Advantages:

- Reduces retransmissions, saving bandwidth.
- Efficient in high-latency networks.

Disadvantages:

- Complex logic for sender and receiver.
- Requires buffer space for out-of-order frames.



Comparison with Go-Back-N ARQ:

Feature	Go-Back-N ARQ	Selective Repeat ARQ
Retransmission	All frames from the error	Only erroneous frames
Efficiency	Lower	Higher
Receiver Buffering	Not required	Required for out-of-order frames
Complexity	Simpler implementation	More complex

Piggybacking:

- **Piggybacking** is a technique used in data communication to optimize the transmission of acknowledgment (ACK) signals by combining them with outgoing data frames.
- Instead of sending a separate acknowledgment frame, the acknowledgment is included in a data frame traveling in the opposite direction.

Advantages:

- **Reduced Overhead:** Fewer frames are sent, saving bandwidth.
- **Improved Efficiency:** Combines two operations (data transmission and acknowledgment) into one.

Disadvantages:

- **Delay in ACKs:** Acknowledgment might be delayed if the receiver has no data to send immediately.
- **Increased Complexity:** Additional logic is required to combine and manage acknowledgments.

Example:

- Sender A sends a frame to Receiver B.
- Instead of sending an acknowledgment (ACK) as a separate frame, Receiver B waits for its turn to send data to Sender A.
- Receiver B includes the ACK in its outgoing data frame to Sender A.

❖ Medium Access Control (MAC) Sublayer :

- The **MAC sublayer** is part of the **Data Link Layer** and plays a key role in managing how devices access and use the shared communication medium (like a network channel).
- It directly interacts with the **Physical Layer** and ensures efficient communication, especially in networks where multiple devices share the same channel.

Functions of MAC Sublayer

1. **Addressing:**
    - Provides a unique hardware address (MAC address) for every device on the network.
2. **Medium Access Control:**
    - Determines how multiple devices share the physical communication medium to avoid collisions and maximize efficiency.
3. **Frame Delivery:**
    - Handles the reliable delivery of frames from one device to another over the same network.

Medium Access Control (MAC) Protocols :

- ➢ **Medium Access Control (MAC)** is a protocol layer that determines how devices on a network gain access to the shared communication medium.
- ➢ It helps manage the timing of data transmission and resolves contention when multiple devices try to transmit simultaneously.
- ➢ MAC protocols ensure efficient and fair use of the channel, minimize collisions, and provide fair access to all devices.

MAC protocols can be broadly categorized into **Random Access** protocols and **Multiple Access** protocols, which are methods used for controlling access to the shared medium.

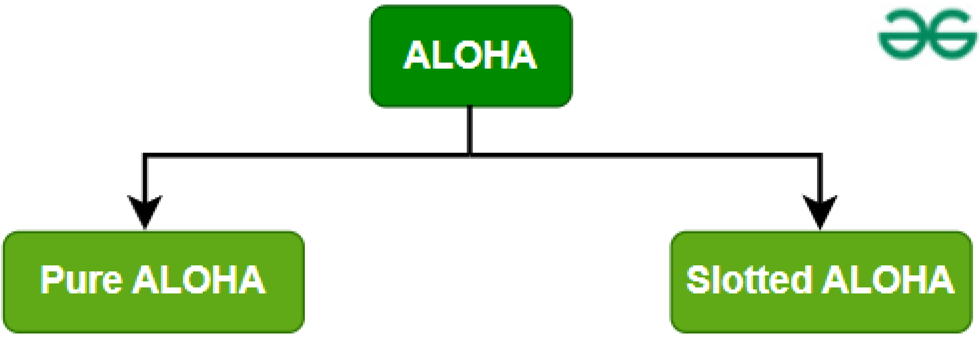
1. Random Access Protocols:

- **Random Access Protocols** allow devices to transmit at any time without coordination, which can lead to **collisions** if multiple devices transmit simultaneously.
- There is no fixed time for sending data
- There is no fixed sequence of stations sending data

The **Random access protocols** are further subdivided as:

❖ ALOHA :

It was designed for wireless LAN but is also applicable for shared medium. In this, multiple stations can transmit data at the same time and can hence lead to collision and data being garbled.



i. **Pure ALOHA**

When a station sends data it waits for an acknowledgement. If the acknowledgement doesn't come within the allotted time then the station waits for a random amount of time called back-off time (Tb) and re-sends the data.

Since different stations wait for different amount of time, the probability of further collision decreases.

Vulnerable Time = 2* Frame transmission time

Throughput = G exp{-2*G}

Maximum throughput = 0.184 for G=0.5

ii. **Slotted ALOHA :**

It is similar to pure aloha, except that we divide time into slots and sending of data is allowed only at the beginning of these slots. If a station misses out the allowed time, it must wait for the next slot. This reduces the probability of collision.

Vulnerable Time = Frame transmission time
Throughput = G exp{-*G}
Maximum throughput = 0.368 for G=1

2. Multiple Access Control (MAC) Protocols :

- Multiple Access Protocols are methods used in computer networks to control how data is transmitted when multiple devices are trying to communicate over the same network.
- These protocols ensure that data packets are sent and received efficiently, without collisions or interference.
- They help manage the network traffic so that all devices can share the communication channel smoothly and effectively.

Some common multiple access protocols:

1. CSMA/CD (Carrier Sense Multiple Access with Collision Detection)

CSMA/CD is commonly used in Ethernet networks to control how devices access the shared communication channel.

Key Features:

- **Carrier Sense**: The device listens to the channel to check if it's idle or busy before transmitting.
- **Multiple Access**: More than one device can access the channel at the same time.
- **Collision Detection**: While transmitting, devices monitor for collisions. If a collision is detected, they stop and retransmit after waiting for a random backoff time.

Steps:

1. The device listens to the channel.
2. If idle, it transmits.
3. While transmitting, the device checks for collisions.
4. If a collision occurs, the device stops and retransmits after a random delay.

2. CDMA/CA (Code Division Multiple Access with Collision Avoidance) :

CDMA/CA is a technique used in systems like cellular and wireless networks to prevent collisions by using unique codes.

Key Features:

- **Code Division**: Multiple devices transmit at the same time but use different codes to encode their data. This allows overlapping signals without interference.
- **Collision Avoidance**: CDMA/CA uses algorithms to minimize the chance of collisions by assigning unique codes to different transmissions.

Steps:

1. Devices are assigned unique codes for simultaneous transmission on the same frequency.
2. The receiver uses the same codes to separate the data from different devices.
3. Collision Avoidance: Algorithms help reduce the chances of simultaneous transmission collisions.